

API Access

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To integrate Defog in your application, you can use the Defog API via either a RESTful API, or a Python client. This page assumes that you have used the [Quick Start](#) guide to extract metadata from Defog, and link your API key to it.

REST API

Generate SQL query, given a text question

ENDPOINT: `https://api.defog.ai/generate_query_chat`

PARAMETERS:

- `api_key` : Your API key. This is used to verify your identity
- `db_type` : The type of database you are querying. This must be one of:
 - `bigquery`
 - `databricks`
 - `mysql`
 - `postgres`
 - `redshift`
 - `snowflake`
 - `sqlite`
 - `sqlserver`
- `question` : The question you want to ask Defog
- `previous_context` : (optional) The previous context of the conversation. This is used to maintain context across multiple questions in a conversation. This is an array of strings in the format `["previous_question", "previous_sql"]`

(more advanced parameters to be added later)

EXAMPLE:

```
curl -X POST https://api.defog.ai/generate_query_chat \
-H "Content-Type: application/json" \
-d '{
  "api_key": "YOUR_API_KEY",
  "db_type": "postgres",
  "question": "What is the average revenue by region?",
}'
```

RESPONSE:

```
{
  "ran_successfully": true,
  "sql": "SELECT region, AVG(revenue) FROM sales GROUP BY region",
  "previous_context": ["What is the average revenue by region?", "SELECT region, AVG
```

The `sql` field contains the SQL query generated by Defog, and the `previous_context` can be used in the next query to maintain context.

Python Client

You can also use Defog's Python client instead of the API. This client library is a wrapper around the API, but also executes the SQL query on your database and returns the results.

To use the Python client, you can install it via pip:

```
pip install defog
```

Here is an example of how you can use the Python client:

```
from defog import Defog
defog = Defog(YOUR_API_KEY, db_type="YOUR_DB_TYPE", db_creds=YOUR_DB_CREDS)
# you only need to do this once. your api key, database credentials, and db_type wil

# if your db_type is postgres or redshift, then db_creds should be {"host": "YOUR_HO
# if your db_type is mysql, then db_creds should be {"host": "YOUR_HOST", "database"
# if your database type is Snowflake, then db_creds should be {"account": "YOUR_ACCO
```

```
# if your database type is Databricks, then db_creds should be {"server_hostname": "  
# if your database type is SQLServer, then db_creds should be {"server": "YOUR_SERVE  
# if your database type is Bigquery, then db_creds should be {"json_key_path": "/pat  
  
# defog = Defog()  
  
defog.run_query(  
    question="What is the average revenue by region?",  
    previous_context=None  
)
```

The `run_query` method generates the SQL query and executes it on your database. The `previous_context` parameter can be used to maintain context across multiple questions in a conversation.

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